

# The Kingman Coalescent: An All- $n$ Genealogy and Branch-Length Atlas

HCS–C215 research certificate

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## Abstract

We treat the partition-valued Kingman coalescent as one mathematical owner. Unit rate per unordered block pair gives a pure-death block count with  $\lambda_k = \binom{k}{2}$ . Partial fractions close every hypoexponential transition, independent holding times close the MRCA law, and a projective coupling gives finite infinite-sample absorption. Scaling by the number of branches identifies total tree length with the maximum of iid  $\text{Exp}(1/2)$  variables, yielding an exact CDF. Markov determinants are not Artin–Mazur zeta functions.

## 1 Partition owner and block chain

For each  $n \geq 1$ , start from the discrete partition of  $[n]$  and merge each unordered pair of current blocks at rate one. The labelled partition process is the Kingman coalescent. Its block count  $K_t$  starts at  $n$  and has

$$\lambda_k = \binom{k}{2} = k(k-1)/2, \quad Q_{k,k-1} = \lambda_k, \quad Q_{k,k} = -\lambda_k,$$

with state 1 absorbing. Conditional on  $k$  blocks, the next pair is uniform. The family is coupled by restricting a partition on  $[n+1]$  to  $[n]$ .

For  $1 \leq j \leq i$  the all- $n$  transition law is hypoexponential:

$$p_{ij}(t) = \left( \prod_{m=j+1}^i \lambda_m \right) \sum_{\ell=j}^i \frac{e^{-\lambda_\ell t}}{\prod_{m=j, m \neq \ell}^i (\lambda_m - \lambda_\ell)}. \quad (1)$$

The empty-product convention gives  $p_{ii}(t) = e^{-\lambda_i t}$  and the  $i = 1$  absorbing row.

The formula can also be read as a semigroup certificate. Its Laplace transform is

$$\int_0^\infty e^{-st} p_{ij}(t) dt = \frac{\prod_{m=j+1}^i \lambda_m}{\prod_{m=j}^i (s + \lambda_m)},$$

and direct multiplication of these triangular kernels gives  $P(t+u) = P(t)P(u)$ . Thus no independently fitted transition table is needed.

## 2 MRCA time and projective limit

Let  $E_k \sim \text{Exp}(\lambda_k)$  be the holding time at level  $k$ . The pair-clock construction and memorylessness make the  $E_k$  independent of one another and of the uniform merger choices. Thus

$$T_n = \sum_{k=2}^n E_k, \quad M_n(s) := \mathbb{E}[e^{-sT_n}] = \prod_{k=2}^n \frac{\lambda_k}{\lambda_k + s}.$$

Consequently,

$$\mathbb{E}[T_n] = \sum_{k=2}^n \frac{1}{\lambda_k} = 2 \left( 1 - \frac{1}{n} \right), \quad \text{Var}(T_n) = \sum_{k=2}^n \frac{1}{\lambda_k^2}. \quad (2)$$

The restriction coupling is essential for the infinite statement. It gives  $T_n \leq T_{n+1}$  on one probability space, hence  $T_n \uparrow T_\infty$  almost surely. Summability of the means and variances implies finite absorption and

$$\mathbb{E}[T_\infty] = 2, \quad \text{Var}(T_\infty) = 4(2\zeta(2) - 3) = \frac{4\pi^2}{3} - 12,$$

with Laplace transform the convergent product obtained by sending  $n \rightarrow \infty$ . This is a projective-coupling assertion, not a claim that independent marginal sums share a sample path.

The monotonicity can be seen directly from the partition restriction: every ancestor of  $[n]$  is also represented in the restricted  $(n+1)$  process, while the extra leaf can only postpone the time at which all labels have coalesced. This coupling argument is separate from the elementary independent-sum calculation used for each fixed  $n$ .

### 3 Total branch length

Define  $L_n = \int_0^{T_n} K_t dt = \sum_{k=2}^n kE_k$ . Since scaling an exponential by  $k$  changes its rate to  $(k-1)/2$ ,

$$\mathbb{E}[e^{-sL_n}] = \prod_{j=1}^{n-1} \frac{j/2}{j/2 + s}, \quad \mathbb{E}[L_n] = 2H_{n-1}, \quad \text{Var}(L_n) = 4H_{n-1}^{(2)}.$$

Let  $Y_1, \dots, Y_{n-1}$  be iid  $\text{Exp}(1/2)$  variables. Their order-statistic spacings, read from the largest backwards, have rates  $(n-1)/2, (n-2)/2, \dots, 1/2$ . Therefore the sum above has the same law as  $\max_i Y_i$ , and the exact CDF is

$$\Pr\{L_n \leq \ell\} = (1 - e^{-\ell/2})^{n-1}, \quad \ell \geq 0. \quad (3)$$

At  $n=1$ ,  $L_1 = T_1 = 0$  and the convention in (3) is the constant one.

For orientation, the first Bell numbers are

$$1, 2, 5, 15, 52, 203, 877, 4140$$

for  $n=1, \dots, 8$ ; they are used only as an independent finite partition ledger, not as an approximation to the infinite process.

### 4 Certificate and source boundary

The evidence payload contains 312 transition rows through  $n=12$ , 12 holding rows, 48 MRCA rows, 60 branch rows, and an independent Bell-number partition ledger through  $n=8$ . The checker recomputes partial fractions, row sums, Chapman–Kolmogorov, moments, the maximum CDF, and the  $n=1$  boundary. SymPy independently verifies the rational transforms and beta-integral behind (3); replay and hostile mutations test reproducibility and schema closure.

The construction is attributed to Kingman (1982), without a priority or novelty claim. It has no intrinsic rational-prime carrier, primitive orbit clock, or arithmetic divisor. The strict Route-A verdict is

$$(\text{AO\_FAIL}, \text{A1\_FAIL}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FAIL}), \quad \text{ROUTE\_A\_REJECTED}.$$

The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. In particular, a finite Markov determinant, trace-log, or Laplace product is not an Artin–Mazur dynamical zeta and is not used as one.

| certificate block               | rows    | independent audit                           |
|---------------------------------|---------|---------------------------------------------|
| transitions ( $n \leq 12$ )     | 312     | partial fractions, sums, Chapman–Kolmogorov |
| MRCA / tree length              | 48 / 60 | moments, products, exact CDF                |
| partition ledger ( $n \leq 8$ ) | 8       | Bell-number enumeration                     |

### References

- [1] J. F. C. Kingman, “The coalescent,” *Stochastic Processes and their Applications* 13, 235–248 (1982), [doi:10.1016/0304-4149\(82\)90011-4](https://doi.org/10.1016/0304-4149(82)90011-4).